137 MHz V-dipole for weather satellites

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Make it yours

Centre frequency 137.5 MHz · V angle 120 ° · Copper wire 1.5 mm² solid-core house wire, ø1.4 mm copper · Insulation Stripped bare (the length is exact) · Mast pipe PVC-U pressure pipe, 40 mm outside, PN16 · Design wind gust 35 m/s gusts (exposed roof or garden) · Mast pipe length 1000 mm · Coax length 3000 mm · Choke Five turns of the coax round the mast

1 Checks

	Check	Result
✓	**Every print fits the printer** As it sits on the bed, turned the way the files come.	`Feed hub: 99 × 104 × 81 on a 220 × 220 × 250 bed`
✓	**Nothing floating** Every member, plate and part touches what holds it.	`all connected`
✓	**No clashes** No member passes through another away from a joint, and no plate through a member it is not notched around.	`none`
✓	**Every piece fits the stock** Cut from standard lengths with a trim and a kerf per cut.	`longest 1000 mm`
✓	**Every piece can be clamped and cut** No piece shorter than 32 mm on its short side.	`shortest 497 mm`
✓	**Wind on mast and antenna head: strength** Yield ÷ peak stress with 51 N of drag in 35 m/s gusts (27 N on the mast, 24 N on the hub, arms, wires, clips and caps, at the top), plus self weight.	`safety factor 2.54 (need 2), worst C (mast)`
✓	**Coax snagged at the hub: strength** Yield ÷ peak stress with 25 N pulled sideways at the hub, plus self weight.	`safety factor 3.99 (need 2), worst C (mast)`
✓	**Every part has a step** Each piece is cut, fitted or bought in a numbered step.	`9 steps`
✓	**Element length from frequency** Measured from the wire's inner end in the hub. End effect for a ø1.4 mm conductor 1.6 %; the PVC arm and clips touching it about 1 %; bare copper. Cut 1 % long and trimmed at the tip in step 8.	`λ/4 at 137.5 MHz = 545 mm; × 0.984 × 0.99 = 531 mm per wire; cut 536 mm`
✓	**V angle** At 120° the horizontal V has its most gain overhead and a usable lobe down to low passes, and its impedance comes close to 50 Ω. Narrower drops the impedance; 180° is a plain dipole with nulls off its ends.	`120° between the elements`
✓	**Feed gap** Small against the wavelength so it does not detune the feed, and wide enough that the two clamps cannot touch (at least 4 mm).	`12 mm between the wire ends (0.55 % of a wavelength); 13 mm between the two sides' washers and wires`
✓	**Feed match (rough estimate)** Estimate R ≈ 73 Ω × sin²(angle ÷ 2) at resonance; the NanoVNA in step 8 gives the real number. Under 2 is plenty for receiving.	`about 55 Ω, SWR 1.09 on 50 Ω coax`
✓	**Choke on the coax** At least 500 Ω of common-mode impedance keeps current off the outside of the shield, so the coax does not become part of the antenna and pick up house noise. Coil from Wheeler's formula; ferrites from the datasheet curve.	`5 turns, about 1.10 µH, 951 Ω at 137.5 MHz`
✓	**Coax loss**	`0.8 dB over 3.00 m of RG58 with its plugs`

	Under 3 dB keeps the weak satellite signal above the receiver noise; use a better cable (or a preamp at the hub) for a longer run.	
✓	**Wire held along the arm** Held at the screw, through the hub channel, by a clip at least every 120 mm and by the tip cap, so the soft wire stays straight on the arm; the tip past the cap carries its own wind and weight against annealed copper (70 MPa).	3 clips per arm every 114 mm, longest run between holds 109 mm; tip 17 mm past the cap, safety factor 96.41, sags 0.0 mm
✓	**Arms in the wind** Each 16 mm conduit arm as a cantilever out of the hub in 35 m/s gusts, with the wire, clips and cap on it and their weight, against PVC's long-term 25 MPa. The design uses the thinner conduit where it holds and the thicker one where it does not.	safety factor 2.44 (need 2); each arm 476 mm out of its socket
✓	**Printed hub fits** Push fits 0.4 mm over the size, at least 3 mm of wall round a socket that is clamped, the wire standing proud by a ring terminal's thickness so the washer clamps both; fits a 200 mm bed.	mast socket ø40.4 mm on ø40 mm pipe; arm sockets ø16.4 mm on ø16 mm conduit, 26 mm deep, 3.5 mm wall; wire channel ø1.8 mm, groove ø1.7 mm with the wire 0.8 mm proud; 99 × 104 on the bed
✓	**Clips and caps fit** The mouth is 75 % of the tube, so the C ring opens 4 mm to pass it: under 2 % strain in PETG or ASA, which springs back. 3 mm ring walls; at least 1.2 mm (three perimeters) over the wire hole.	clip bore ø16.2 mm snug on ø16 mm, mouth 12 mm, springs open 1.4 % to snap on; cap socket ø16.3 mm; wire hole ø1.8 mm for ø1.4 mm copper, 2 mm wall over it

2 Materials

942 × 341 × 1034 mm · 1.0 kg · 5 pieces · 10 prints, about 142 g, 12.5 h · 7.5 h of work · ~$51 · Frequency: 137.5 MHz · Element: 531 mm · V angle: 120° · Tip to tip: 941 mm · Arms: 16 mm PVC conduit · Clips: 6 · Hub height: 1.03 m

Stock

Section	Pieces	Cut length	Buy	Mass	Cost
16 × 1.4 rigid PVC electrical conduit	2	1.0 m	1 × 3.00 m	0.1 kg	$3.00
1.5 mm² solid copper house wire (ø1.4 conductor)	2	1.1 m	2 × 1.00 m	0.0 kg	$1.20
40 × 3 PVC-U pressure pipe, PN16	1	1.0 m	1 × 3.00 m	0.5 kg	$14

3D prints

Part	Settings	Qty	Filament	Print	Step	Cost	Files
Feed hub	PETG · 0.2 mm layers · 4 walls · 40% infill · no supports	1	121 g	10.1 h	1	$2.66	
Pocket lid	PETG · 0.2 mm layers · 4 walls · 40% infill · no supports	1	6 g	0.5 h	1	$0.14	
Wire clip	PETG · 0.2 mm layers · 4 walls · 60% infill · no supports	6	7 g	1.2 h	5	$0.15	
Tip cap	PETG · 0.2 mm layers · 4 walls · 60% infill · no supports	2	8 g	0.7 h	5	$0.18	

Filament and time are estimates from the part's volume; your slicer has the real numbers. The .stl is the part at this size, in millimetres; the .scad is its OpenSCAD source.

Parts to buy

Part	Spec	Qty	Step	Cost
Heat-set insert, M3	M3, short (about 4 long), for a 4.0 hole	6	1	$0.60
SMA female bulkhead jack, solder cup	panel mount, 50 Ω, with nut and lock washer, for panels up to 4 thick	1	2	$2.00
Solder tag to fit the SMA jack thread	tinned brass ground lug	1	2	$0.20
Ring terminal, M3	tinned copper, for 0.5 to 1.5 square wire, solder or crimp	2	2	$0.20
Hook-up wire	stranded, silicone insulated, 0.5 square	1	2	$0.50
Grub screw, M3 × 8, cup point, stainless	into the inserts under the sockets, onto the arms	2	3	$0.20
Screw, M3 × 8, pan head, stainless	clamps each wire end into its insert in the pocket floor	2	4	$0.20
Penny washer, M3 × 12, stainless	3.2 hole, 12 outside, about 1 thick	2	4	$0.20
Bolt, M4 × 60, stainless, with nyloc nut and two washers	through the hub socket and the mast	1	6	$0.40
Coax, RG58, SMA male to SMA male	3.0 metre assembly (or the plug your receiver needs at the far end)	1	7	$9.60
Cable ties, UV-stable black	200 long	9	7	$0.45
Screw, M3 × 8, pan head, stainless	lid into the heat-set inserts	2	9	$0.20
Mast clamp (U-bolt with saddle)	M6 U-bolt for a 40 pipe, with a saddle for your post or wall bracket	2	9	$6.00

Consumables

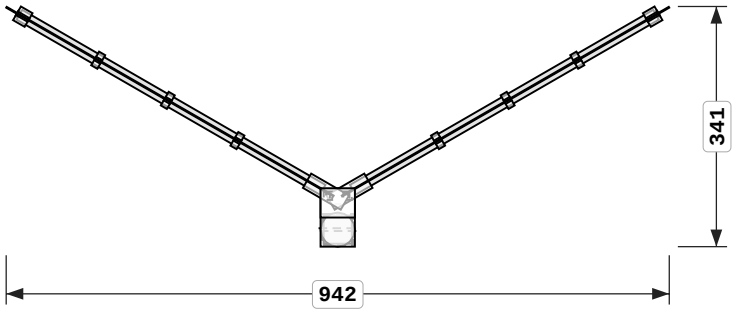
- Solder, rosin core · a little
- Self-amalgamating tape · 1 roll
- PVC electrical tape, black · 1 roll
- Neutral-cure silicone sealant · 1 small tube

Tools

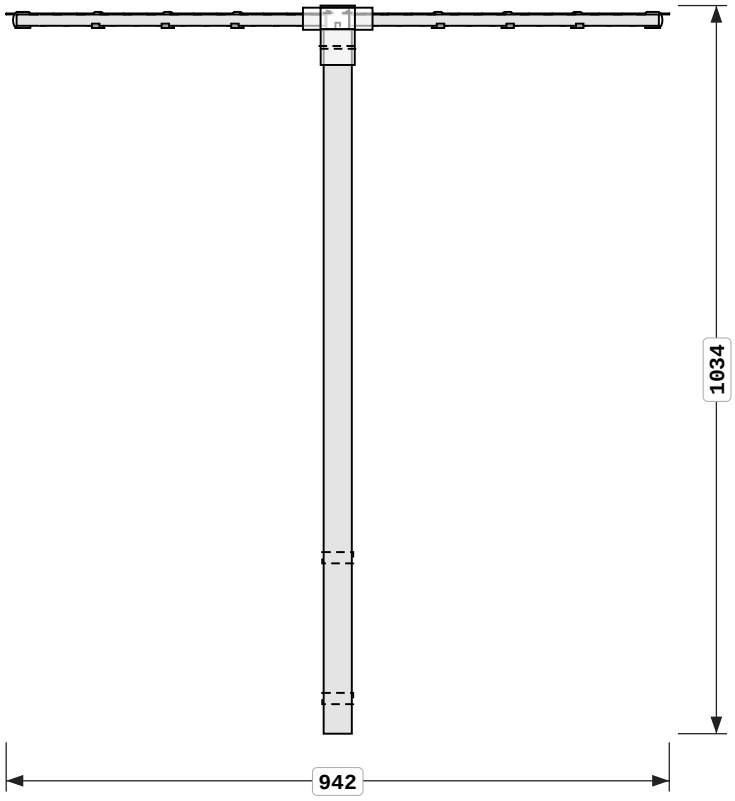
- 3D printer and the filament named with each print
- Soldering iron (with a heat-set insert tip if you have one)
- Fine-tooth saw, file and deburring tool
- Drill and a 4.3 mm bit
- Vise, pliers, side cutters, wire strippers and a sharp knife
- Calipers and tape measure
- Multimeter
- NanoVNA or antenna analyser
- Spanner for the SMA nut, screwdrivers, hex key for the grub screws
- Spirit level and compass

3 Drawings

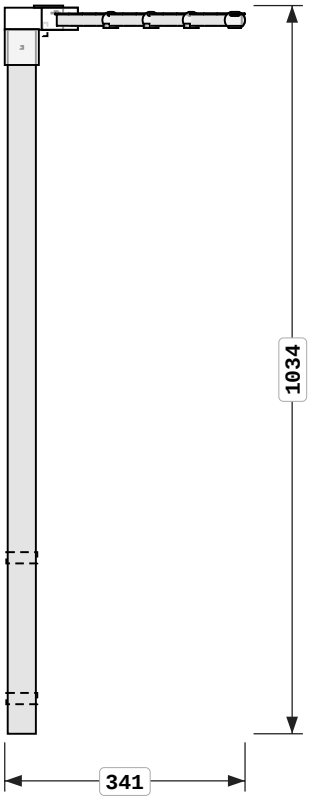
Top



Front



Side (from the right)



4 Legend

Units	millimetres, kilograms, degrees
Datum	x east from the apex of the V (the feed point), y north (the V opens toward north), z up from the bottom end of the mast
Piece marks	A, B, C… one letter per different piece; x4 after it means cut four the same. Mirror images share a mark.
Cut marks	From one square end, mark each edge at the two numbers, join the marks around the section, cut on the lines. Angles are for reference.
Edges	named by where they face on the finished build (top-south, bottom-east…)
Joints	nothing is welded: parts clamp, as each step says
Long point	the overall length of the piece, tip to tip
Dimensions	jig drawings and joint tables give centre lines (℄), where the members' centres cross; the general arrangement gives outside sizes
Tolerance	cut ±1 mm unless a step says otherwise

5 Order

Start the hub print, then wire the jack and cut the arms while it runs; clamp the wires, clip them to the arms, fit the mast, tune by trimming the tips, then seal and mount.

Fabricate

1. Print the hub and the lid 1 h
2. Wire the SMA jack 0.5 h
3. Arms 0.5 h
4. Element wires: cut, strip and clamp 1 h
5. Clips and tip caps 0.5 h
6. Mast and hub 1 h

Assemble

7. Coax and coil choke 0.5 h

Tune and install

8. Tune with a NanoVNA 1 h
9. Seal, mount and point it 1.5 h

6 Steps

1 Print the hub and the lid

1 h

- [ ] 1. Print the hub, then the lid (files below). Queue the 6 clips and 2 tip caps from step 5. PETG or ASA, not PLA.
- [ ] 2. Start steps 2 and 3 while it prints.
- [ ] 3. Clear strings from the sockets, channels and holes. A scrap of wire must slide through each channel.
- [ ] 4. Melt an M3 heat-set insert into each 4 mm hole, flush: two in the pocket, two under the arm sockets, two behind the pocket.

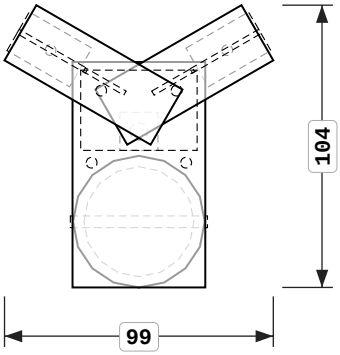
✓ Conduit and mast push into their sockets by hand. Inserts flush ±0.5 mm.

! A tight socket: open it with a file. Never force the tube: the hub splits along its layers.

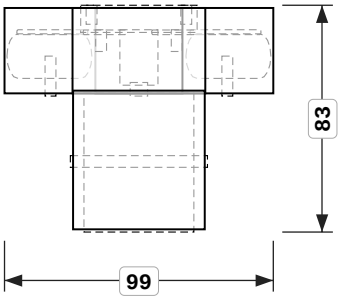
Print: Feed hub

PETG · 0.2 mm layers · 4 walls · 40% infill · no supports · about 121 g, 10.1 h each · print it top face down on the bed, mast socket pointing up (the files come turned this way); no supports: the pocket floor and the tops of the arm bores bridge

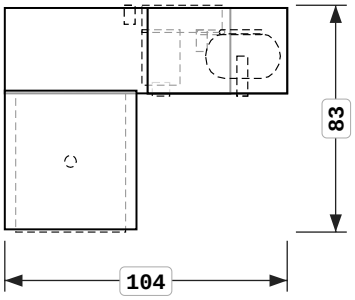
Feed hub: Top



Feed hub: Front



Feed hub: Side (from the right)



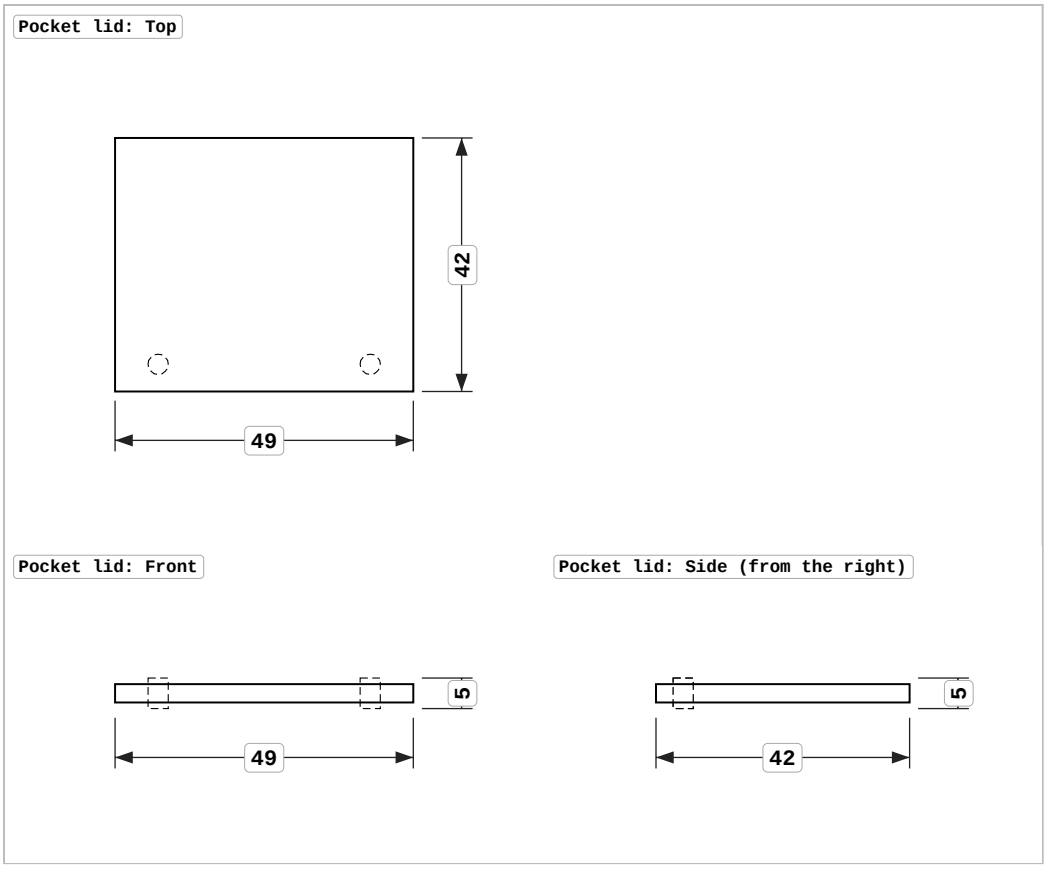
Cut	Size	Axis	From (x, y, z)	Depth
Hole	ø40.4	z	49.4, 24.2, -1	51
Hole	ø4.3	x	24.2, 24.2, 25	50.4
Pocket	42.7 × 29.5 × 10		49.4, 65.2, 77.4	
Pocket	14 × 14 × 20.4		49.4, 57.4, 63.2	
Hole	ø6.4	z	49.4, 57.4, 49	5
Hole	ø16.4	angled	70.8, 80.9, 63.7	26.6
Hole	ø16.4	angled	28, 80.9, 63.7	26.6
Hole	ø1.8	angled	68.7, 79.7, 72.5	29.1
Hole	ø1.8	angled	30.2, 79.7, 72.5	29.1
Hole	ø1.7	angled	54.6, 71.5, 72.5	17.3
Hole	ø1.7	angled	44.3, 71.5, 72.5	17.3
Hole	ø4	z	63.3, 72.4, 65.4	8
Hole	ø4	z	35.6, 72.4, 65.4	8
Hole	ø4	z	81.9, 87.3, 49	14.7
Hole	ø4	z	16.9, 87.3, 49	14.7
Hole	ø4	z	32.1, 45.9, 75.4	7
Hole	ø4	z	66.8, 45.9, 75.4	7

Dashed on the drawing. Positions from the part's lower-left-bottom corner as drawn; the files come turned for printing (+z face on the bed). On the bed it is 98.8 × 103.8 × 81.4 mm.

! Melt an M3 heat-set insert into each of the six small holes: two in the pocket floor, two under the arm sockets, two on the top face behind the pocket.

Print: Pocket lid

PETG · 0.2 mm layers · 4 walls · 40% infill · no supports · about 6 g, 0.5 h each · print it flat on the bed



Cut	Size	Axis	From (x, y, z)	Depth
Hole	ø3.3	z	7, 4.5, -1	5
Hole	ø3.3	z	41.7, 4.5, -1	5

Dashed on the drawing. Positions from the part's lower-left-bottom corner as drawn. On the bed it is 48.7 × 41.5 × 3 mm.

! Screwed down over a bead of neutral-cure silicone after tuning.

Parts

- Heat-set insert, M3 ×6 M3, short (about 4 long), for a 4.0 hole

2 Wire the SMA jack

0.5 h

- [ ] 1. Two hook-up wire tails, 50 mm. Strip 4 mm each end. An M3 ring terminal on one end of each.
- [ ] 2. Solder tag over the jack thread. One tail to the centre cup (east), one to the tag (west).
- [ ] 3. Jack down through the well behind the pocket. Lock washer and nut from under; hold the body, tighten.
- [ ] 4. Centre tail to the east insert, tag tail to the west one.

✓ Multimeter: centre pin to body open; each pin to its ring terminal under 1 Ω.

! Solder before the jack goes in: the insulator and the PETG soften under a slow iron.

Parts

- SMA female bulkhead jack, solder cup x1 panel mount, 50 Ω, with nut and lock washer, for panels up to 4 thick
- Solder tag to fit the SMA jack thread x1 tinned brass ground lug
- Ring terminal, M3 ×2 tinned copper, for 0.5 to 1.5 square wire, solder or crimp
- Hook-up wire x1 stranded, silicone insulated, 0.5 square

3 Arms

0.5 h

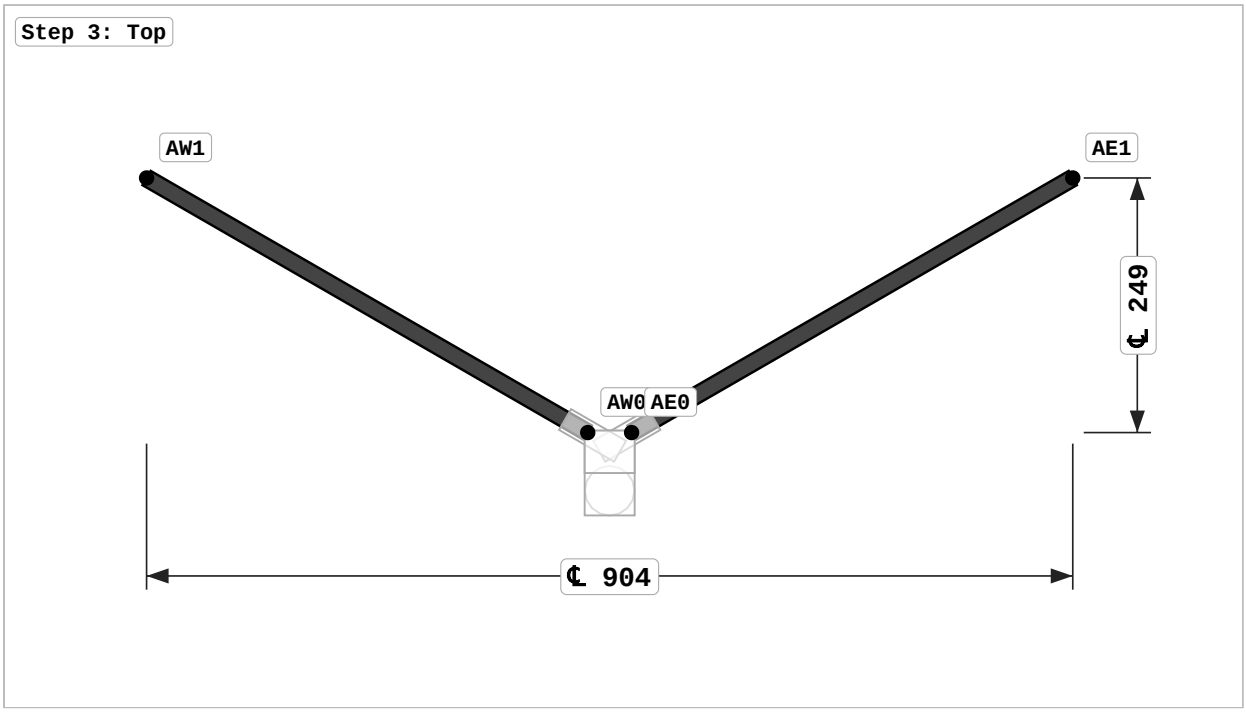
- [ ] 1. Cut two arms (A) from 16 mm conduit, square. Deburr.
- [ ] 2. Push each into its socket until it bottoms.
- [ ] 3. Grub screw in each insert until it bears, then a quarter turn.

✓ Tip to tip 904 ±5 mm. Arms do not turn or pull out. Tips level ±3 mm on a flat table.

Parts

- Grub screw, M3 × 8, cup point, stainless x2 into the inserts under the sockets, onto the arms

Jig



Joint positions

Point	x	y	z
AW0	-21	12	1014
AE0	21	12	1014
AW1	-452	261	1014
AE1	452	261	1014

Pieces: A x2

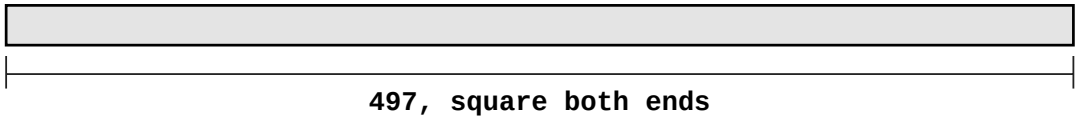
A x2

497 mm

Arm · Conduit 16 · step 3

long point

`arm-E, arm-W`



Left end: free end · Right end: free end

4 Element wires: cut, strip and clamp

1 h

- [ ] 1. Straighten the 1.5 mm² wire: one end in a vise, pull the other until it lies dead straight. Cut two wires (B).
- [ ] 2. Strip all the insulation: slit it lengthwise, peel it off. Do not nick the copper.
- [ ] 3. Feed each inner end through its arm channel into the pocket groove.
- [ ] 4. Inner ends 12 mm apart.
- [ ] 5. On each insert: ring terminal, penny washer over terminal and wire, M3 × 8 screw. Tighten until the wire flattens.

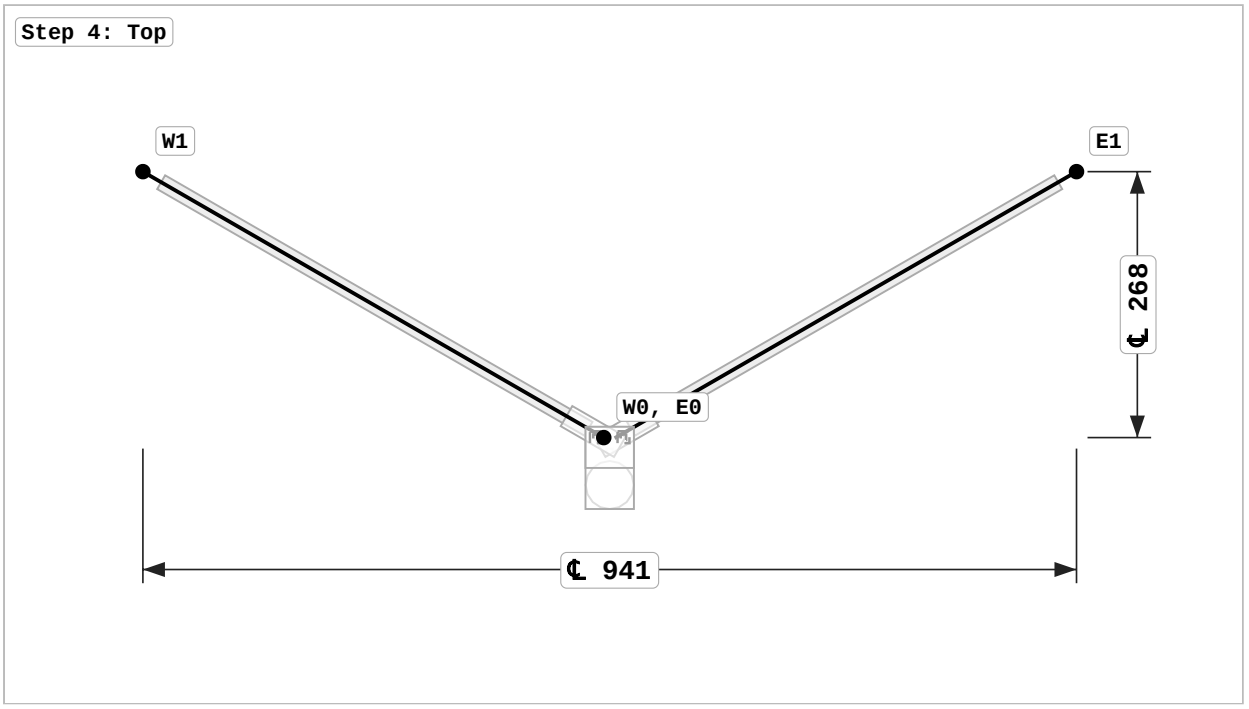
✓ Gap 12 ±1 mm. Wires stay put when tugged. Each wire to its pin under 1 Ω; pin to body still open.

! Set the gap before you tighten. The wires are 5 mm long on purpose: step 8 trims them.

Parts

- Screw, M3 × 8, pan head, stainless x2 clamps each wire end into its insert in the pocket floor
- Penny washer, M3 × 12, stainless x2 3.2 hole, 12 outside, about 1 thick

Jig



Joint positions

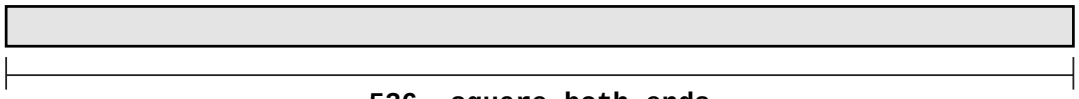
Point	x	y	z
W0	-6	3	1022
E0	6	3	1022
W1	-470	272	1022
E1	470	272	1022

Pieces: B x2

B x2 — **536 mm**

Element wire · Cu wire 1.5 · step 4 — long point

`wire-E, wire-W`



536, square both ends

Left end: free end · Right end: free end

5 Clips and tip caps

0.5 h

- [ ] 1. Print 6 clips and 2 tip caps if not done. A clip must click onto an offcut and grip.
- [ ] 2. From the tip end: 3 clips onto each wire, mouths down, then a tip cap.
- [ ] 3. Cap over the arm end until it bottoms. A dab of neutral-cure silicone.
- [ ] 4. Mark the clip centres (table). Snap each clip onto the arm on its mark.
- [ ] 5. Pull the wire straight. Tip 17 mm past the cap.

✓ No kinks. Clips on their marks ±5 mm. Tips 17 ±3 mm past the caps. Tip to tip 941 ±5 mm.

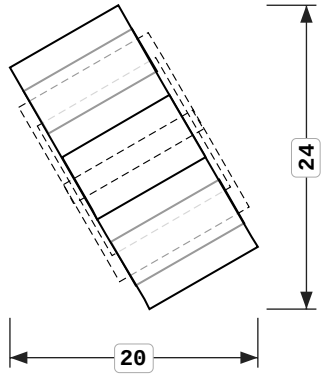
Clip centres along each arm

Clip	From the hub arm end
Clip 1	114
Clip 2	228
Clip 3	342

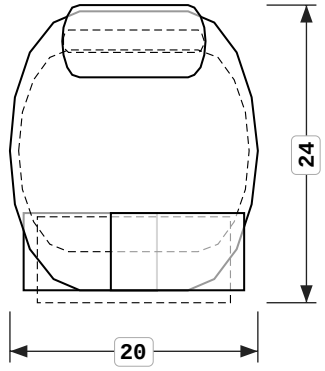
Print: Wire clip x6

PETG · 0.2 mm layers · 4 walls · 60% infill · no supports · about 1 g, 0.2 h each · print it as modelled: lying on its side, the open mouth down, standing on its two flat feet; no supports (the top of the bore bridges)

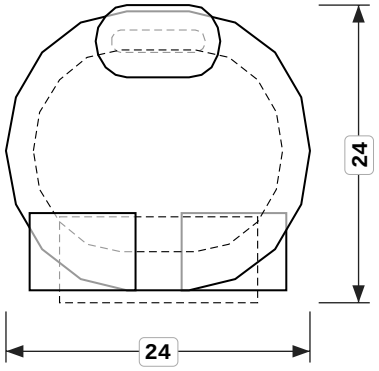
Wire clip: Top



Wire clip: Front



Wire clip: Side (from the right)



Cut	Size	Axis	From (x, y, z)	Depth
Hole	ø16.2	angled	4.7, 9.2, 11.2	12
Hole	ø1.8	angled	4.7, 9.2, 20	12
Cut	prism			

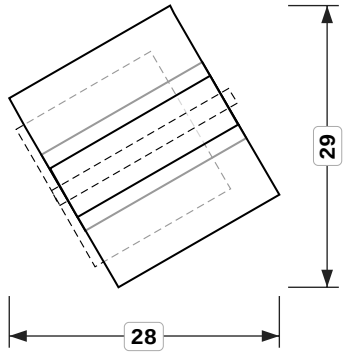
Dashed on the drawing. Positions from the part's lower-left-bottom corner as drawn. On the bed it is 19.9 × 24.4 × 22.9 mm.

! Thread 3 onto each wire before the tip cap; snap onto the arm from above.

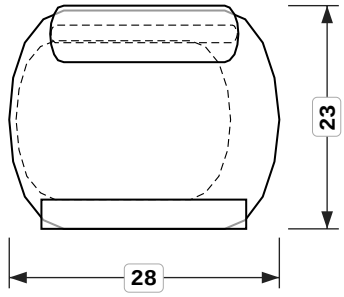
Print: Tip cap x2

PETG · 0.2 mm layers · 4 walls · 60% infill · no supports · about 4 g, 0.3 h each · print it as modelled: lying on its flat foot, the socket and end wall standing up; no supports (the top of the socket bridges)

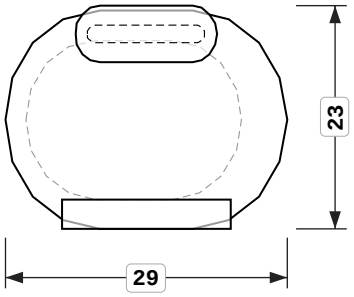
Tip cap: Top



Tip cap: Front



Tip cap: Side (from the right)



Cut	Size	Axis	From (x, y, z)	Depth
Hole	ø16.3	angled	4.7, 9.2, 11.2	16
Hole	ø1.8	angled	4.7, 9.2, 20	21

Dashed on the drawing. Positions from the part's lower-left-bottom corner as drawn. On the bed it is 27.7 × 28.9 × 22.9 mm.

! Pushed over the end of the arm; the wire comes out through the end wall.

6 Mast and hub

1 h

- [ ] 1. Cut the mast (C) from 40 mm PVC pipe, square. Deburr.
- [ ] 2. Drill 4.3 mm straight through, 25 mm below the top.
- [ ] 3. Hub onto the mast until it bottoms. Holes in line. M4 bolt, washers, nyloc nut.
- [ ] 4. Mark the clamps 50 and 250 mm up from the bottom.

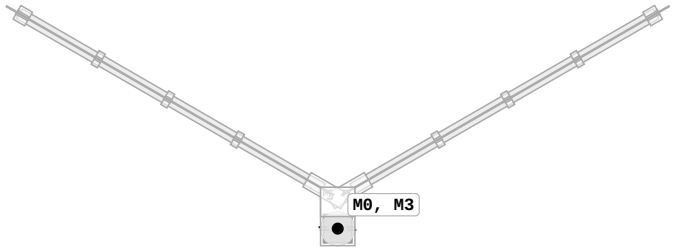
✓ Mast ±1 mm of its length. Hub does not turn or lift. Arms level with the mast plumb.

Parts

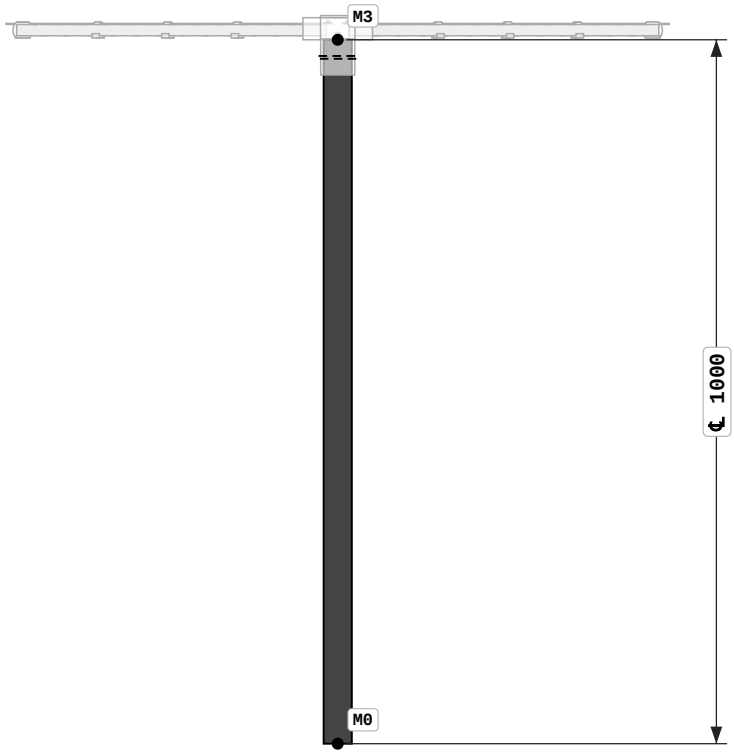
- Bolt, M4 × 60, stainless, with nyloc nut and two washers x1 through the hub socket and the mast

Jig

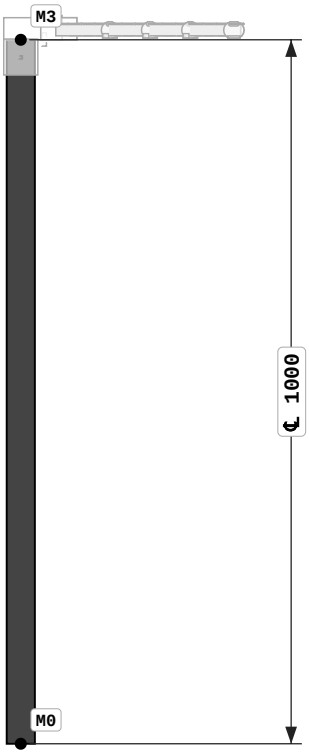
Step 6: Top



Step 6: Front



Step 6: Side (from the right)



Joint positions

Point	x	y	z
M0	0	-44	0
M3	0	-44	1000

Pieces: C x1

C x1

1000 mm

Mast · PVC 40×3 · step 6

long point

mast



Left end: free end · Right end: free end

7 Coax and coil choke

0.5 h

- [ ] 1. Coax SMA plug onto the jack: finger tight, then an eighth of a turn.
- [ ] 2. Self-amalgamating tape over the joint, half lapped. PVC tape over it.
- [ ] 3. 5 tight turns of coax round the mast right under the hub (about 25 mm). Tape each end.
- [ ] 4. Coax down the mast, a cable tie every 200 mm. A drip loop where it leaves.

✓ 5 turns, tight, starting 50 mm or less under the hub. Nothing slides.

Parts

- Coax, RG58, SMA male to SMA male x1 3.0 metre assembly (or the plug your receiver needs at the far end)
- Cable ties, UV-stable black x9 200 long

8 Tune with a NanoVNA

1 h

- [ ] 1. Mast upright in the open, clear of walls, metal and people.
- [ ] 2. Calibrate the NanoVNA (open, short, load). Connect the antenna coax.
- [ ] 3. Sweep 120 to 160 MHz. Aim: lowest SWR at 137.5 MHz ±1 MHz. As cut, expect it low.
- [ ] 4. Low: snip both tips equally, about 3.9 mm per MHz. Sweep again.
- [ ] 5. High: fine if SWR at 137.1 and 137.9 MHz is under 2; else cut a new pair 3.9 mm longer per MHz.
- [ ] 6. No NanoVNA: a licensed operator can read SWR at 144 and 146 MHz. Low at 144, rising to 146.

SWR minimum ±1 MHz of 137.5 MHz. Under 2 from 137.1 to 137.9 MHz. Both tips equal ±1 mm.

Trim a little at a time: a millimetre cannot be put back. Bend nothing at the tips.

9 Seal, mount and point it

1.5 h

- [ ] 1. Fill the pocket with neutral-cure silicone (not acetic). A bead at each channel. Lid on, two M3 × 8 screws.
- [ ] 2. Two U-bolt clamps on the marks, 50 and 250 mm up. Both tight.
- [ ] 3. V open to true north or south.
- [ ] 4. Mast plumb, arms level, as much sky as possible, away from other antennas.
- [ ] 5. Coax to the receiver, drip loop below the entry.

Open within 10° of north or south. Arms level within 3°. No movement when shaken at the hub.

Both clamps tight on a post that does not move: the wind check counts on them.

Parts

- Screw, M3 × 8, pan head, stainless x2 lid into the heat-set inserts
- Mast clamp (U-bolt with saddle) x2 M6 U-bolt for a 40 pipe, with a saddle for your post or wall bracket

7 Cut list

Mark	Qty	Section	Long point	Left end	Right end	Step
A Arm	2	Conduit 16	497	square · free end	square · free end	3
B Element wire	2	Cu wire 1.5	536	square · free end	square · free end	4
C Mast	1	PVC 40×3	1000	square · free end	square · free end	6

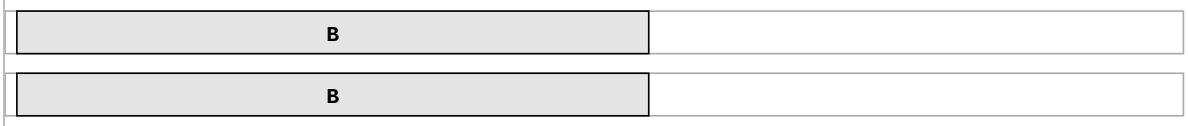
Cutting plan

Each bar is one stock length; cut the pieces left to right.

Conduit 16 · 1 × 3.00 m lengths · kerf 2



Cu wire 1.5 · 2 × 1.00 m lengths · kerf 2



PVC 40×3 · 1 × 3.00 m lengths · kerf 2



137 MHz V-dipole for weather satellites · v2. Generated by the blueprint skill. Check every dimension against your own stock before you cut, wear eye and hand protection, and load-test the build gently before trusting it.